Practice Set B

Class 12 Mathematics, Chapter 8: Application of Integrals

Maximum marks: **53** **20** questions

Name _____ Date _____ Score _____ / 53

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1 — 8 marks

1 MCQ, Board-pattern — 1 mark

The area of the region $\{(x, y) : |x| + |y| \leq 1\}$ is:

(A) 1 sq unit **(B)** 4 sq units **(C)** $\frac{1}{2}$ sq unit **(D)** 2 sq units

2 MCQ, Board-pattern — 1 mark

If the area of the region in the first quadrant enclosed by the circle $x^2 + y^2 = r^2$ and the coordinate axes is 4π sq units, then r is equal to:

(A) 2 **(B)** 4 **(C)** 8 **(D)** 16

3 MCQ, Board-pattern — 1 mark

If the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$ $(a > 0)$ is 10π sq units, then a is equal to:

(A) 25 **(B)** $\frac{5}{2}$ **(C)** 5 **(D)** 10

4 MCQ, Board-pattern — 1 mark

The area of the region bounded by the curve $y = \cos x$ and the x-axis between $x = 0$ and $x = \pi$ is:

(A) 2 sq units **(B)** 4 sq units **(C)** 3 sq units **(D)** 1 sq unit

5 MCQ, Board-pattern — 1 mark

The area of the region bounded by the line $x = 2y + 3$, the y-axis and the lines $y = 1$ and $y = -1$ is:

(A) 4 sq units **(B)** $\frac{3}{2}$ sq units **(C)** 6 sq units **(D)** 8 sq units

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** The area of the region bounded by the curve $y = \sqrt{x}$, the x-axis and the line $x = 4$ is $\frac{16}{3}$ sq units. **Reason (R):** The area of the region bounded by the curve $y = \sqrt{x}$, the y-axis and the line $y = 2$ is also $\frac{16}{3}$ sq units. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, PYQ-style The area between $x = y^2$ and $x = 4$ is divided into two equal parts by the line $x = k$. Find the value of k.	2 marks

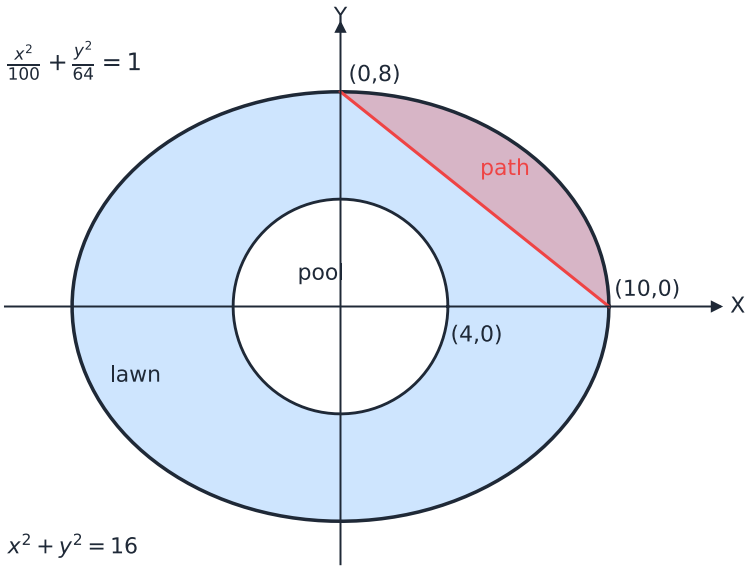
Level 2

18 marks

8	Short answer, Board-pattern The curve $x = at^2$, $y = 2at$ $(a > 0)$ is a parabola. Find the area of the region enclosed by this parabola and the ordinates (vertical lines) through its points where $t = 1$ and $t = 2$.	3 marks
9	Short answer, Board-pattern Using integration, find the area of the region bounded by the parabola $y^2 = 4x$, the y-axis and the lines $y = -2$ and $y = 4$.	3 marks
10	Short answer, CBSE SQP 2025-26 Sketch the graph of $y = \|x + 1\|$. Evaluate $\int_{-4}^{2} \|x + 1\|\, dx$. What does the value of this integral represent on the graph?	3 marks
11	Short answer, Board-pattern Using integration, find the area of the region in the first quadrant bounded by the ellipse $\frac{x^2}{4} + \frac{y^2}{16} = 1$, the y-axis, the x-axis and the line $y = 2$.	2 marks
12	Short answer, Board-pattern Using integration, find the area of the region bounded by the curve $y = \sqrt{36 - x^2}$, the x-axis and the lines $x = -3$ and $x = 3\sqrt{3}$.	3 marks
13	Short answer, Board-pattern Find the area of the region bounded by the curve $y = \sin 2x$, the x-axis and the lines $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$.	2 marks
14	Short answer, Board-pattern Find the area of the region bounded by the curve $y = x^3$, the line $y = x + 6$ and the line $x = 0$.	2 marks

Q No.	Question	Marks
Level 3		**27 marks**
15	Long answer, Board-pattern Using integration, find the area of the region $\{(x, y) : \lvert x\rvert \leq y \leq \sqrt{8 - x^2}\}$.	5 marks
16	Long answer, Board-pattern Using integration, compute the area of the region bounded by the lines $x + 2y = 2$, $y - x = 1$ and $2x + y = 7$.	5 marks
17	Long answer, Board-pattern Find the area of the region bounded by the parabola $y^2 = 2x$ and the straight line $x - y = 4$.	5 marks
18	Long answer, PYQ-style Using integration, find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Hence, find the area enclosed by the ellipse $16x^2 + 9y^2 = 144$.	5 marks
19	Short answer, Board-pattern Using integration, find the area of the region bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$.	3 marks

Q No.	Question	Marks
20	Case study, Board-pattern **Case Study:** A housing society has an elliptical lawn whose boundary is the ellipse $\frac{x^2}{100}+\frac{y^2}{64}=1$, with its centre at the origin (units in metres). At the centre there is a circular fountain pool with boundary $x^2+y^2=16$. Grass covers the lawn everywhere outside the pool. A straight footpath of negligible width joins the points $(10,0)$ and $(0,8)$ on the boundary. Based on this information, answer the following questions using integration.	4 marks



(i) Find the area enclosed by the fountain pool. 1 mark

(ii) Find the area enclosed by the elliptical boundary of the lawn. 1 mark

(iii)(a) Find the area of the part of the lawn in the first quadrant that lies between the footpath and the boundary. 2 marks

OR

(iii)(b) Find the area of the grassed part of the lawn lying in the first quadrant. 2 marks

Answer key

Chapter 8: Application of Integrals, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(D) 2 sq units
2	(B) 4
3	(C) 5
4	(A) 2 sq units
5	(C) 6 sq units
6	(C) A is true, but R is false
7	$k = 4^{2/3}$ (that is, $\sqrt[3]{16}$)

Level 2

8	$\frac{56a^2}{3}$ sq units
9	6 sq units
10	9; it is the area bounded by $y = \lvert x+1 \rvert$, the x-axis and the lines $x = -4$ and $x = 2$
11	$\left(\sqrt{3} + \frac{2\pi}{3}\right)$ sq units
12	$9\left(\sqrt{3} + \pi\right)$ sq units
13	1 sq unit
14	10 sq units

Level 3

15	2π sq units
16	6 sq units
17	18 sq units
18	πab sq units; 12π sq units
19	$\frac{9}{8}$ sq units
20	(i) $16\pi\ \mathrm{m}^2$ (ii) $80\pi\ \mathrm{m}^2$ (iii)(a) $(20\pi - 40)\ \mathrm{m}^2$ OR (iii)(b) $16\pi\ \mathrm{m}^2$